Domain-wall erosion and the complete periodic atlas of cyclic majority rule 232

Route-A source-local dynamics study

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Abstract

We study the synchronous radius-one majority cellular automaton on every labelled binary cycle. Passing to domain walls gives the exact Boolean law $w_i' = w_i(1 \oplus w_{i-1} \oplus w_{i+1})$: each finite wall block erodes by one site at either end per tick. This yields a complete periodic classification: all non-alternating states enter the fixed language in at most $\lfloor (n-1)/2 \rfloor$ steps, while the two alternating words at even length form the unique nontrivial two-cycle. Fixed states are counted for every n by a four-state de Bruijn trace, $\# \operatorname{Fix}(F_n) = L_n + 2\cos(n\pi/3)$. Parity-twisted run matrices give exact transient-depth populations. These are finite-state, source-local results; they do not supply arithmetic data or a Hilbert–Pólya operator.

1 Frozen dynamics

For $n \geq 1$, let $X_n = \{0,1\}^n$ with indices modulo n, and define

$$(F_n x)_i = \mathbf{1}\{x_{i-1} + x_i + x_{i+1} \geq 2\}. \tag{1}$$

The clock is one simultaneous update. Words are labelled: rotations and complements are not quotiented. The small cases $n = 1, 2$ are retained as degenerate faces, while the wall proof below is uniform. The distinction between a transient and a recurrent word is part of the object definition, rather than an inference from a finite simulation.

2 The wall reduction

Set $w_i = x_i \oplus x_{i+1}$. A site changes exactly when it is isolated, that is, when $w_{i-1} = w_i = 1$. Taking the xor of two neighbouring updates in (1) gives the local identity

$$w_i' = w_i(1 \oplus w_{i-1} \oplus w_{i+1}) \qquad \text{in } \mathbb{F}_2. \tag{2}$$

For completeness, the eight neighbourhoods are checked directly in the accompanying certificate. If a block 1^K is bounded by zeros, only its two endpoint walls disappear. Hence

$$K(t) = \max\{K - 2t,\, K \bmod 2\}. \tag{3}$$

Blocks stay separated until one vanishes. The exceptional all-one wall word exists only for even n and is the alternating pair.

Lemma 1 (entry time)**.** *Let K_j be the finite wall-block lengths of x. If x is not alternating, then*

$$\tau(x) = \max_j \lfloor K_j/2 \rfloor \leq \left\lfloor \frac{n-1}{2} \right\rfloor. \tag{4}$$

The bound is sharp for a single zero wall followed by a maximal alternating block.

3 Periodic states

When every wall block has length at most one, no symbol is isolated and the state is fixed. Conversely, a fixed state cannot contain 010 or 101. The all-one wall word is the only case not covered by the erosion lemma.

Theorem 2 (complete periodic classification)**.** *For every n,* $\mathrm{Per}(F_n)$ *consists of the fixed words avoiding* 010 *and* 101*, together with one temporal two-cycle when n is even, represented by* $0101\cdots$ *and* $1010\cdots$*. There are no temporal periods greater than two.*

Proof. Equation (3) sends every wall word containing a zero to a word with no adjacent ones, hence to a fixed state. The only wall word containing no zero is 1^n; it is compatible with the xor constraint exactly when n is even, and (1) exchanges its two lifts. This also excludes every other cycle. □

4 Exact fixed counts

The fixed language is a cyclic finite-type language with forbidden triples 010 and 101. On pair states $00, 01, 10, 11$, its de Bruijn matrix is

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}. \tag{5}$$

Closed walks give $a_n = \mathrm{tr}(M^n)$. A determinant calculation factors its characteristic polynomial as

$$\chi_M(\lambda) = (\lambda^2 - \lambda - 1)(\lambda^2 - \lambda + 1). \tag{6}$$

Writing $L_0 = 2, L_1 = 1, L_{n+2} = L_{n+1} + L_n$ and $c_0 = 2, c_1 = 1, c_{n+2} = c_{n+1} - c_n$, we obtain

$$\#\,\mathrm{Fix}(F_n) = a_n = L_n + c_n = L_n + 2\cos(n\pi/3). \tag{7}$$

The first values are $2, 2, 2, 6, 12, 20, 30, 46, 74, 122, 200, 324$.

5 Transient-depth enumeration

For a run bound m, let B_m have states $0, \dots, m$ and entries $B_m[r, 0] = 1$, $B_m[r, r+1] = 1$ for $r < m$. The parity-twisted matrix B_m^- changes the latter entries to -1. Then

$$E_{n,m} = \frac{\mathrm{tr}(B_m^n) + \mathrm{tr}((B_m^-)^n)}{2} \tag{8}$$

counts cyclic wall words of even parity and maximum run at most m. Each wall word lifts to two labelled binary words. Therefore differences of $2E_{n,2t+1}$ and $2E_{n,2t-1}$ give the exact number entering the fixed set at time t; for even n the all-one wall word is added separately as the two-cycle. This separates a genuine transient census from the recurrent periodic core.

6 Independent certificate and Route-A boundary

The integer producer records 64 fixed-count rows, 216 run-matrix rows, and an exhaustive state census through $n = 14$. A producer-independent checker passes 1,849 assertions; an independent SymPy program passes 569 identities; a clean byte replay passes; and all 40 hostile mutations are rejected. The source commit, evaluator hash, fixed epoch, evidence payload, PDF, and file ledger are closed in the release manifest.

The strict Route-A tuple is `(A0_FAIL,A1_PASS_ANALYTIC,A2_FAIL,A3_FAIL,A4_FORMAL_HINT)`, with overall verdict `ROUTE_A_REJECTED`; Route B is disabled. No target prime/zero table, arithmetic local datum, Euler factor, root number, automorphy statement, target divisor or functional equation, or Hilbert–Pólya operator is claimed.

References

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[2] A. L. Toom, “Stable and attractive trajectories in multicomponent systems,” in *Multicomponent Random Systems* (1980), 549–575, doi:10.1007/978-1.4613-3044-2_19.